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Biological quality assessment of different generations of the parasitoid, *telenomus busseolae* Gahan (Hymenoptera: Scelionidae) reared on eggs of the sugarcane stem borer *sesamia nonagrioides* Lefèbvre (Lepidoptera: Noctuidae)

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Abstract

Background The present study evaluated the biological parameters (parasitism rate, fertility, emergence rate, adult longevity, and offspring sex ratio) of seven generations (F_0 , F_1 , F_3 , F_5 , F_7 , F_8 , and F_{10}) of the parasitoid wasp *Telenomus busseolae* Gahan (Hymenoptera: Scelionidae) reared on eggs of the sugarcane stem borer *Sesamia nonagrioides* Lefèbvre (Lepidoptera: Noctuidae) under laboratory conditions.

Results A host colony was established from field-collected *S. nonagrioides*, and fresh eggs were provided to *T. busseolae* females from generations F_0 (field-collected), F_1 , F_3 , F_5 , F_7 , F_8 , and F_{10} ; 30 female wasps per generation were randomly selected for the parameter recording. Significant declines were observed across generations: parasitism rate decreased from 85.2% (F_0) to 61.3% (F_{10}), fertility from 42.1 to 25.7 parasitized eggs/female, emergence rate from 92.3 to 71.8%, female longevity from 9.2 to 5.8 days, representing an approximately 37% reduction. The offspring sex ratio shifted from 75% female (significantly female-biased) to 45% female, which did not differ significantly from a 1:1 ratio.

Conclusions The results demonstrate a gradual decline in the biological quality of *T. busseolae* under laboratory rearing. Consequently, periodic introduction of field populations, optimization of rearing diets, and stringent environmental control are recommended to sustain the efficacy of *T. busseolae* in biological control programs.

Keywords *Telenomus busseolae*, Parasitism rate, Biological parameters, Generational decline, Rearing, Biocontrol

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Background

As a cornerstone of industrial agriculture in tropical and subtropical regions worldwide, sugarcane (*Saccharum officinarum* L. hybrids) fulfills a vital role, supplying approximately 70% of global sugar (Hunsigi 2012). The crop's economic significance extends far beyond sugar; it holds remarkable potential for generating a diverse portfolio of industrial outputs, including bagasse, straw, molasses, filter cake, and ethanol (Paturau 1989). Iran's sugarcane industry is predominantly centered in Khuzestan province, a region whose very name ("the land of sugar") reflects its agricultural identity as the country's primary cultivation area (Hamdi 2016).

Sesamia spp. (Lepidoptera: Noctuidae), known as the sugarcane stem borers, ranks as a primary threat to sugarcane cultivation in Khuzestan province, with additional impacts on maize and rice (Agah-Manesh et al. 2021). Larval damage manifests as "dead heart" in young plants and structural compromise or death in mature plants through stem boring. Consequently, larval infestations cause substantial reductions in sugarcane juice volume and quality. The pest exhibits four yearly generations in this region. Its nocturnal adult females deposit eggs specifically behind the leaf sheaths of living sugarcane plants (Khanjani 2009).

Chemical control proves ineffective against *Sesamia nonagrioides* Lefebvre (Lepidoptera: Noctuidae) because its larvae are protected from chemical exposure by dint of feeding within the sugarcane stems (Halabian et al. 2013). This limitation has led to the adoption of inoculative releases of the egg parasitoid *Telenomus busseolae* Gahan (Hymenoptera: Scelionidae) as the principal control measure in Khuzestan's sugarcane agroecosystems (Nikpay and Goebel 2016). However, this parasitoid wasp does not consistently suppress pest populations below damaging levels across a range of sugarcane cultivars (Halabian et al. 2013).

Given the vital role of this wasp in controlling the insect pest, rearing this parasitoid under laboratory conditions is essential and essential component of the biological control program. However, the biological quality of wasps in successive generations may deteriorate due to rearing conditions. Declines in parasitism rate, longevity, fecundity, and optimal sex ratio in later generations can reduce field efficacy. Therefore, rigorous assessment of *T. busseolae* quality indicators across generations and comparison with wild populations is critical for optimizing biological control strategies. Studies on various egg parasitoid wasps, including *Telenomus remus* Nixon (Hymenoptera: Platygasteridae) (Pomari-Fernandes et al. 2015) and *Trichogramma brassicae* Bezdenko (Hymenoptera: Trichogrammatidae) (Ghaemmaghami et al. 2021a; b.), have shown that rearing conditions significantly impact the biological quality and biological control

efficacy of individuals reared over successive generations. Specifically, without rigorous quality assessment and the selection of robust isolines, parasitoids frequently exhibit reduced parasitism rates, fertility, and optimal sex ratios, which directly compromise sustained field efficacy against target pests (Chen et al. 2023). Therefore, rigorous assessment of *T. busseolae* quality indicators across generations and comparison with wild populations is critical for optimizing biological control strategies.

The present study aimed to assess the impact of successive rearing on the biological characteristics and efficacy of biological control by *T. busseolae*, with the goal of applying the findings to the development of the biological control of sugarcane stem borers.

Methods

To establish a laboratory colony, *S. nonagrioides* larvae were collected from infested sugarcane fields in the Imam Khomeini (31.801993, 48.767391) and Haft Tappeh (32.086303, 48.402481) agro-industrial complexes. Infested stems were identified by desiccated apical buds or larval entry holes containing darkened frass. The stems were then dissected to extract 4th and 5th larval instars, thereby reducing the duration of the pupal stage before colony establishment. Larvae were reared in cylindrical plastic containers (25 cm height × 16 cm diameter) and provided with fresh sugarcane stems under controlled laboratory conditions. After pupation, individuals were sexed based on genital slit morphology and isolated in separate cylindrical containers (5 cm height × 10 cm diameter). For each oviposition cage, 10 pairs of newly emerged adults (sexed as pupae) were introduced. Mating was confirmed by observation of copulation, and only egg masses laid after confirmed mating were used in experiments. All developmental stages were maintained in a germinator at 28 ± 1 °C, $60 \pm 5\%$ relative humidity, and a 14:10 (Light: Dark) photoperiod. Throughout the experiments, the same host colony was used as the source of eggs. While strict environmental and dietary controls were applied to maintain host quality, wild-type moths were not periodically introduced. We acknowledge this as a study limitation, as any untracked decrease in host colony quality or potential artificial genetic bottleneck effects from the initial field collection could have compounded the observed decline in parasitoid biological parameters.

The parasitoid, *T. busseolae* was reared on its natural host, *S. nonagrioides* eggs. The initial culture was established from parasitized *S. nonagrioides* egg masses collected from the sugarcane fields. In the laboratory, host eggs (< 24 h old) were exposed to newly emerged parasitoid species. For oviposition, *S. nonagrioides* eggs were glued onto 15 × 2 cm strips of radiographic polyester film using a 14% honey-water solution and presented to the

parasitoid. The egg-bearing strips and parasitoids were maintained in cylindrical glass tubes (17 cm length $\times$ 3 cm diameter), which were sealed with cotton plugs. Adult wasps were provided with filter paper strips saturated with a 20% honey-water solution. All tubes were maintained in a germinator at 28 ± 1 °C, $60 \pm 5\%$ relative humidity, and a 16:8 (Light: Dark) photoperiod.

Biological quality assessment of *Telenomus busseolae*

Quality control parameters of *T. busseolae* were evaluated across seven laboratory-reared generations (F0, F1, F3, F5, F7, F8, and F10), where F0 represented the field-collected generation. For each generation, 30 randomly selected female wasps (mated for 24 h post-emergence) were individually housed in glass tubes. Each female was provided daily with 30 fresh host eggs (<24 h old) until her death. The number of parasitized eggs was recorded daily, and the total parasitism rate was calculated as the proportion of eggs parasitized during the first 24 h of exposure (to standardize across generations). Fecundity was calculated as the total number of eggs parasitized over the female's lifetime.

Statistical analysis

Data normality and homogeneity of variances were checked using Shapiro–Wilk and Levene's tests, respectively. For normally distributed data, one-way ANOVA followed by Tukey's HSD test was applied. For proportional data (emergence rate), data were arcsine-square-root transformed before ANOVA. Sex ratio data were analyzed using a generalized linear model (GLM) with a binomial error distribution and logit link function (logistic ANOVA), with generation as the fixed factor, following the recommendations of Wilson and Hardy (2002). While arrhenotoky provides mated females with the physiological mechanism to control sex allocation, it is the population structure and mating system that primarily drive the evolutionary expectation of female-biased sex ratios in parasitoid wasps (Hardy and Boulton 2024);

therefore, deviations from a 1:1 ratio were interpreted within this ecological context. All analyses were performed using SPSS version 26 (IBM Corp., Armonk, NY, USA). Each experiment was repeated 30 times.

Results

Results indicated that the F0 generation (field-collected) exhibited the highest parasitism rate (Table 1). A gradual decline in parasitism rate was observed across laboratory-reared generations (F1 to F10). Tukey's HSD test revealed significant differences in parasitism rates among generations ($df=6, 203; F=31.2; P<0.05$). The lowest parasitism rate was observed in F10 (61.3%), which was 28.1% lower than F0 (85.2%).

Female wasp fertility was assessed by counting the number of eggs parasitized by each female. There was significant difference between wasp fertility in different generations ($df=6, 203; F=27.6; P<0.05$) (Table 1). The early generations (F0 and F1) exhibited higher mean fertility with 42.1 and 39.5 eggs per female, respectively, compared to subsequent generations. Furthermore, a decline in fertility was observed across successive generations. The lowest fecundity was observed in F10 (25.7 eggs per parasitoid), which was 39.0% lower than F0 (42.1 eggs). These findings indicated a gradual decline in reproductive capacity in laboratory-reared generations.

Results indicated a significant gradual decrease in the lifespan of female *T. busseolae* ($df=6, 203; F=19.4; P<0.001$), with mean lifespan decreasing from 9.2 days in the F₀ generation to 5.8 days in the F₁₀ generation, representing a 37.0% reduction. (Table 1).

The emergence rate was calculated by dividing the number of parasitized eggs with exit holes relative to the total number of parasitized eggs. The emergence rate was the highest in F0 generation and decreased across subsequent generations. The difference in emergence rates between generations was statistically significant ($df=6, 203; F=24.8; P<0.001$). Emergence rate of the parasitoid decreased by 22.2% from F0 (92.3%) to F10 (71.8%) (Table 1).

Logistic ANOVA (GLM with binomial distribution) revealed a significant overall effect of generation on offspring sex ratio (Wald $\chi^2 = 18.45, df=6, P=0.005$). Pairwise comparisons against the 1:1 expectation showed that the sex ratio was significantly female-biased in the F0 ($P=0.006$) and F1 ($P=0.033$) generations. From F3 onward, the offspring sex ratio did not differ significantly from 1:1 ($P>0.05$), indicating a progressive loss of female-biased allocation in later laboratory generations (Table 2).

Table 1 Parasitism rate, fertility, female lifespan, and emergence percentage of different generations of *Telenomus busseolae* on *Sesamia nonagrioides* eggs

Generation	Parasitism rate (%)	Mean Number of parasitized eggs $\pm$ SE	Female wasp lifespan $\pm$ SE (day)	Emergence (%)
F0	85.2a*	42.1 $\pm$ 0.65 a	9.2 $\pm$ 0.25 a	92.3a
F1	80.4b	39.5 $\pm$ 0.72 b	8.7 $\pm$ 0.27 b	89.7 b
F3	76.8c	36.8 $\pm$ 0.78 c	8.1 $\pm$ 0.29 c	86.2 c
F5	72.5d	33.2 $\pm$ 0.85	7.5 $\pm$ 0.31 d	82.5 d
F7	68.9f	30.6 $\pm$ 0.92	6.9 $\pm$ 0.34 e	78.4 e
F8	65.7 g	28.4 $\pm$ 0.96 f	6.3 $\pm$ 0.36 f	75.1 f
F10	61.3 h	25.7 $\pm$ 1.03 g	5.8 $\pm$ 0.38 g	71.8 g

Means followed by the same letter within a column are non-significantly different (Tukey's test, $P<0.05$)

Table 2 Sex ratio of *Telenomus busseolae* reared on *Sesamia nonagrioides* eggs across different generations

Generation	N	Female sex ratio (%)	SE	Wald χ^2 (df=1)	P-value	Significance
F0	30	76.7 (23/30)	0.080	7.59	0.006	Female-biased (s)
F1	30	70.0 (21/30)	0.084	4.52	0.033	Female-biased (s)
F3	30	66.7 (20/30)	0.087	3.20	0.073	Marginally ns
F5	30	60.0 (18/30)	0.089	1.18	0.277	ns
F7	30	56.7 (17/30)	0.091	0.53	0.467	ns
F8	30	50.0 (15/30)	0.091	0.00	1.000	ns
F10	30	46.7 (14/30)	0.091	0.13	0.715	ns

Overall model: Wald $\chi^2 = 18.45$, df=6, $P=0.005$. P-values represent pairwise comparisons of each generation's sex ratio against the 1:1 expectation using logistic regression (GLM, binomial family, logit link) with Bonferroni correction. N=30 per generation. ns: not significantly different from 1:1 ($P>0.05$)

Discussion

Biological fitness parameters (reproductive output, lifespan, emergence rate, and sex ratio) and parasitism rates of *T. busseolae* declined significantly across successive generations under laboratory rearing conditions. This progressive decline could significantly compromise the effectiveness of *T. busseolae* as a biological control against *S. nonagrioides*.

Preserving the quality of natural enemies across multiple generations is critical, as prolonged mass rearing under stable insectary conditions may reduce their field efficacy (Bertin et al. 2017). Large-scale parasitoid production for augmentative biological control demands significant labor resources while risking deterioration in agent quality over successive generations (Parra 2010). This degradation manifests as reduced environmental tolerance and altered host specificity, ultimately compromising pest management efficacy (Sørensen et al. 2012). Consequently, rigorous quality control throughout rearing and deployment progress is essential for biological control program success. The observed reduction in parasitism rate and fertility can be attributed to multiple interacting factors, including adaptation to artificial laboratory conditions, reduced host-searching efficiency, and nutritional deficiencies associated with continuous rearing on the same host species (Chen et al. 2023). Recent studies on *T. remus* and *Trichogramma* spp. have demonstrated that without rigorous quality assessment and the selection of robust isolines, parasitoids frequently exhibit reduced parasitism rates and fertility, which directly compromise their long-term efficacy under field conditions (de Paula et al. 2023).

Furthermore, the significant decrease in emergence rate and female longevity observed in later generations indicates a loss of physiological vigor and developmental stability. This deterioration is often a consequence of inbreeding depression and metabolic changes induced by the artificial rearing environment (Ghaemmamghami et al. 2021). Reduced emergence rates may result from increased embryonic mortality due to the expression of deleterious recessive alleles, while shortened longevity reflects a decline in stress tolerance and overall metabolic

efficiency (Wang et al. 2022). These findings are consistent with broader evidence showing that mass-reared egg parasitoids frequently exhibit reduced survival and flight activity over successive generations, necessitating continuous monitoring of these vital quality indicators (Chen et al. 2023).

Laboratory rearing of *T. remus* has been reported to reduce foraging and flight capacity, compromising field efficacy in biological control programs (Naranjo-Guevara 2020). These impairments are consistent with broader evidence that mass-reared egg parasitoids frequently exhibit reduced parasitism rates, fertility, host searching efficiency, and flight activity (de Paula et al. 2023). The decline in fecundity and parasitism rate may result from multiple interacting factors, including reduced foraging ability, inbreeding depression, host quality variation, and potential adaptation to laboratory conditions that may compromise field-relevant traits.

Fecundity is a critical determinant mass production costs and serves as a key quality indicator for parasitoids, however, it often declines during prolonged laboratory rearing. In this study, *T. busseolae* exhibited a reduction in fecundity over 10 generations, mirroring with declines reported in *T. embryophagum* (Sadat et al. 2021) and *T. brassicae* reared on lepidopteran eggs (Ghaemmamghami et al. 2021a). Notably, the biological quality of *T. busseolae*, as reflected by fecundity and parasitism rate, declined within only 10 generations, a deterioration that occurred disproportionately faster than in other parasitoid wasps such as *T. brassicae*, which maintained quality for over 40 generations under similar rearing conditions (Ghaemmamghami et al. 2021b), or specific robust isolines of *T. remus* that require careful selection to avoid quality loss (de Lacerda et al. 2023). This suggests heightened sensitivity of *T. busseolae* to artificial rearing conditions and inbreeding, necessitating colony restoration through wild-type supplementation.

Our findings align with Naranjo-Guevara et al. (2020), who demonstrated that prolonged laboratory rearing can select for parasitoids with reduced fecundity, aggression, and flight activity compared to wild counterparts. Wild

individuals consistently exhibit higher parasitism performance and faster population growth rates.

A major concern in the mass-rearing natural enemies is inbreeding, defined as the mating between genetically related individuals, which triggers inbreeding depression. This evolutionary process manifests as reduced offspring fitness. In diploid females, inbreeding increases homozygosity, thereby increasing the likelihood of deleterious recessive deleterious recessive alleles. In haploid males, deleterious alleles are inherently and always expressed, collectively compromise survival, reproduction, and adaptive traits (Charlesworth and Willis 2009). Recent genomic and ecological studies confirm that prolonged mass rearing of parasitoid wasps can rapidly reduce genetic diversity and host-searching efficiency unless these traits are actively monitored and managed (Li et al. 2024; de Melo et al. 2024). The significant generational effect on sex ratio confirmed by logistic ANOVA further supports the progressive loss of female-biased allocation as a consequence of inbreeding. Although the specific molecular mechanism of sex determination in *T. busseolae* has not been explicitly characterized in the literature, this rapid deterioration and sex ratio distortion are highly consistent with the effects of complementary sex determination (CSD). Under CSD, inbreeding increases the production of homozygous diploid males, which are often sterile or exhibit reduced fitness, thereby imposing a severe genetic load on mass-reared colonies (Cook and Crozier 1995). In arrhenotokous parasitoids, sex ratio manipulation is primarily controlled by females through differential fertilization of eggs; however, this ability may be disrupted by inbreeding depression (Russell and Stouthamer 2010). The male-biased shift observed in *T. busseolae* is likely a consequence of increased homozygosity, which exposes deleterious recessive alleles that disproportionately affect female offspring survival or impair female reproductive capacity, including sperm storage and utilization (Charlesworth and Willis 2009). Recent studies further emphasize that such sex ratio distortions in mass-reared colonies can substantially reduce population growth potential and biological control efficacy, underscoring the need for periodic genetic supplementation with wild populations (Li et al. 2023).

Concern about the decline in the female sex ratio is another major challenge in using *T. busseolae* against the sugarcane stem borer. As emphasized in biological control literature, a shift in the sex ratio towards male-bias and a reduction in females (the individuals directly responsible for parasitizing the host pest) will lead to a severe decrease in field efficacy (Russell and Stouthamer 2010). Offspring sex ratios in parasitoid species are influenced by multiple factors (e.g., parental age, temperature, host traits, density), primarily through female-mediated control of egg fertilization and differential survival).

Overall, the rapid deterioration of *T. busseolae* biological quality within only 10 generations—a rate disproportionately faster than in other parasitoid species such as *T. brassicae*, which maintained quality for over 40 generations under similar rearing conditions (Ghaemmaghami et al. 2021a)—suggests a heightened sensitivity of this species to artificial rearing and inbreeding. The combined declines in parasitism, fertility, emergence, longevity, and sex ratio directly threaten the operational viability of *T. busseolae* in augmentative biological control programs. To mitigate these effects, rigorous quality control protocols, including the periodic introduction of wild genetic material, optimization of rearing diets, and the potential use of iso-female line selection, are essential for sustaining the efficacy of this parasitoid against *S. nonagrioides* in sugarcane agroecosystems (Fortes et al. 2023).

Although the arcsine-square-root transformation followed by ANOVA provided robust statistical comparisons in this study, future research could further enhance the analysis of proportional data (e.g., parasitism rate, emergence rate) by employing logistic regression (GLM with binomial distribution). The models directly account for the binomial nature of the response variable and may provide a more appropriate framework for analyzing proportion-based outcomes. (Hardy and Smith 2023).

Conclusions

The findings indicate that the biological quality of *T. busseolae* may deteriorate over successive generations (F_0 – F_{10}) under laboratory rearing conditions. Critical parameters, including parasitism rate, fecundity, emergence rate, adult longevity, and female-biased sex ratio, appeared to decline progressively, with the F_{10} generation exhibiting reductions of 28–39% compared to the wild-derived F_0 . This relatively rapid degradation (within 10 generations) suggests that *T. busseolae* may be particularly sensitive to prolonged laboratory rearing and the inbreeding effects, although direct comparisons to other parasitoid species require standardized experimental protocols. Notably, these findings contrast with previous studies in which key quality parameters remained relatively stable across multiple generations. Recommended interventions include periodic introduction of wild genetic material, optimization of rearing diets, and strict control of rearing conditions. These measures may be critical for sustaining the operational viability of *T. busseolae* as a biological control agent and for ensuring consistent pest suppression in sugarcane agro-systems.

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Author contributions

M.S. and F.Y. conceived and designed the study. M.S. performed the experiments and collected the data. N.Z.-S. and A.R. contributed to data analysis and interpretation. F.Y. wrote the manuscript. All authors read and approved the final manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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